

Compressible Tasks in Green Data Centers as Grid-Forming Support Assets

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Abstract—Renewable microgrids can help data centers cope with the increasing demand for cloud-based services, but they pay the price of the double uncertainty of workload and renewable resource availability. The mismatch between demand and renewable supply prevents microgrid-based data centers from achieving full autonomy, requiring connection to the main grid. However, while imports from the main grid could increase environmental impact, the massive export of excess renewables, when injected into the grid, can cause serious instability in the system. This work addresses both issues using *compressible tasks*, whose concave utility functions allow for the adaptation to the available resources, reducing or increasing the task size with negligible impact on the quality provided to the user. By absorbing surplus renewable energy, they help maintain frequency stability in the main grid, thereby supporting grid-forming operations. We cast this problem as an optimization problem that dynamically adapts the task size based on renewable availability. We devised *Online-REATA*, an online heuristic able to manage tasks arrivals on the fly. Tested in a live video transcoding case study and compared to a solution without task compression, *Online-REATA* achieves the same average QoE and self-sufficiency, as well as higher self-consumption, while reducing storage requirements by up to 35.3%.

Index Terms—green data center; sustainability; grid-forming.

I. INTRODUCTION

In recent years, increasing attention is devoted to the substantial rise in Data Centers (DCs) energy consumption resulting from the expanding demand for cloud-based services [1]. This expansion is expected to continue in the coming years, amplifying the urgency of identifying sustainable energy solutions [2], [3]. Green Data Centers (GDCs) with microgrids for on-site renewable production are an effective response, aiming for self-sufficiency and reduced dependence on the main grid. However, they need to address the mismatch between energy demand and production, due to the inherent intermittency of renewable energy.

On one hand, GDCs often have to import energy from the main grid (semi-autonomous green data centers), typically with a carbon intensity that is country-dependent and rarely fully renewable. On the other hand, in case of renewable energy surplus, data centers connected to the main grid often send extra renewable energy back to the grid, affecting its stability. This is because variable renewable generation from

inverter-based resources (IBRs), lacking the inertia of conventional rotating machines operating at 50 Hz, can cause frequency deviations that threaten grid stability and potentially trigger cascading failures and blackouts [4].

To mitigate these effects, modern power systems increasingly rely on *grid-forming control*, which refers to the capability of enforcing voltage and frequency in the main grid [5]. However, frequency stability remains challenging in renewable-rich grids due to fast and unpredictable power variations [6]. In this context, it becomes necessary to enhance the autonomy of data centers from the main grid by minimizing energy imports and exports. Two metrics are generally used to capture the two aspects: (i) the *Self-Sufficiency Rate (SSR)*, which measures the share of demand covered by renewable energy, indicating the system's capacity to operate independently, and (ii) the *Self-Consumption Rate (SCR)*, which measures the share of renewables absorbed locally, showing how much renewable energy the system can use. We consider SSR and SCR as defined in reference [7], which counts the energy stored in the battery as part of the renewable supply.

In this work, we propose to support grid-forming control by adapting workloads to renewable availability through compressible tasks, whose size can be flexibly adjusted to absorb renewable fluctuations. *Compressible tasks* [8], [9] are computational jobs where the utility function is concave: compressing them (i.e., reducing their length, and resource usage) slightly degrades their utility, while increasing them enables absorbing surplus energy that would otherwise destabilize the grid. Then, our main contributions can be listed as:

- 1) We model compressible task allocation within green data centers as an *offline* optimization problem. We then develop an *online* heuristic algorithm that jointly addresses renewable surplus and shortage by leveraging compressible tasks as *grid-forming-supportive loads*, capable of enhancing frequency stability.
- 2) We conduct a case study on live video streaming, adjusting the transcoding task size to reduce interaction between the microgrid and the main grid. Our results demonstrate that our proposal can increase self-consumption and self-sufficiency while reducing battery size.

The paper is organized as follows. Section II reviews the related work, Section III presents the model, Section IV

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introduces the case study, Section V reports the evaluation and results. Section VI summarizes our conclusion.

II. RELATED WORK

The uncertain nature of renewable energy production can result in deficits and surpluses [7], which can be compensated for by importing and exporting energy to and from the main grid, respectively. Imports occur in data centers equipped with renewable energy systems but also connected to the main grid, importing energy whenever renewable production is insufficient. Scientific literature largely focuses on reducing grid imports to lower carbon emissions [10], while the issue of exports remains less explored. When there is a surplus of renewable energy generated, this energy needs to be exported to the main grid. However, the exported energy derived from inverter-based resources (IBRs) including renewable energy generators (wind, solar) and batteries, can destabilize the grid. Unlike conventional synchronous generators, IBRs lack inertia and provide limited short-circuit current, making frequency deviations more likely and potentially leading to cascading outages [4]. Reference [7] proposed focusing on self-consumption and peak shaving to reduce energy exports and imports, yet presenting these two objectives as mutually exclusive. **In our proposal, we argue that the real objective should be to reduce overall interaction with the main grid, reducing dependence during imports and alleviating stability issues during exports.**

An approach that can jointly mitigate both shortages and surpluses is the usage of an Energy Storage System (ESS). In [3], the authors compare a grid-dependent system with different autonomous DC configurations, showing that combining a Battery Energy Storage System (BESS) with a Hydrogen Energy Storage System (HESS) offers the best performance toward full autonomy. Full autonomy remains cost-prohibitive compared to European energy prices. The leap from 95% to 100% renewable coverage requires a disproportionate infrastructure increase (five times the 0-95% requirement), making it economically and environmentally unsustainable [11]. Moreover, the environmental impact of battery production reduces the benefits, suggesting hybrid solutions as a more realistic alternative. **In this work, we show that compressible tasks can lower the required battery size, while supporting grid-forming by reducing grid interactions.**

Previous studies have proposed shifting the execution of tasks over time or space to match renewable availability—either delaying tasks or migrating them across data centers [10]–[12]. However, both approaches present major drawbacks. Spatial shifting increases the carbon footprint due to VM migration energy costs, especially when network power isn't green, and its efficacy is limited by latency constraints, bandwidth, and destination resource availability [12]. Moreover, spatial distribution causes loss of global system knowledge and communication overhead [10]. Temporal shifting requires extra server capacity, leading to higher embedded carbon [11]. Finally, combining both strategies may increase total energy use, because more servers must stay powered on

for longer periods [12]. **In this work, rather than shifting demand in time or space, our approach adjusts the duration (size) of workloads to better match available generation from the local microgrid.**

The closest related study is [10], which explores quality degradation to reduce grid imports in distributed DCs (some are grid-connected, other are autonomous and equipped with Photovoltaic (PV) and/or batteries). It proposes load negotiation, consolidation, and resolution switching to a lower video quality under resource scarcity. These three strategies combined yield a +7.83% SCR increase and 35.7% footprint reduction. **In contrast to our work, authors in [10] do not provide a quantification of the degradation, neither they envisage quality enhancements to manage the surplus.** Finally, the authors of [10] address excess generation in reference [12], by modeling interconnected DCs that exchange surplus energy through the main grid (imports from one can be offset by simultaneous exports from another). **In contrast to us, this mitigates the issue only economically, without resolving the physical limitations of large renewable injections into the grid.**

Summing up, existing approaches mainly address imports or exports separately, without exploring the potential of leveraging *compressible tasks* to jointly mitigate both issues.

III. RENEWABLE ENERGY AWARE TASK ALLOCATION

Let a compressible task j be characterized by a concave utility function $U_j(x_j)$, which mirrors the Chebyshev approximation error model and its saturation behavior [13]. It is defined as:

$$U_j(f_j = F \cdot x_j) = 1 - \tau_j \cdot \exp(-\theta_j \cdot F \cdot x_j) \quad j \in \mathcal{J} \quad (1)$$

The utility depends on the CPU usage x_j allocated to the task j (x_j can be larger than one, if more than a CPU core is used). Considering F the core frequency, the number of cycles per second used by each task j is $f_j = F \cdot x_j$. The coefficients θ_j and τ_j are task-specific parameters and capture task efficiency. Notations in this section are summarized in Table I. Time is modeled at two granularities: at the coarse scale, discrete time slots $t \in \mathcal{T}$ represent the periods over which the renewable energy input R_t may vary. At the fine scale, intervals $i \in \mathcal{I}$ are defined between consecutive task arrivals or completions, during which the active task set is constant.

For each task j , Eq. (2) defines $X_j^{\min}$ as the minimum number of cores required to ensure its baseline utility $U^{\min}$.

$$X_j^{\min} = U_j^{-1}(U^{\min}) = -\frac{\ln\left(\frac{1-U^{\min}}{\tau_j}\right)}{\theta_j \cdot F} \quad (2)$$

$$P_{ti}^{\min} = \sum_{j \in \mathcal{J}_i} P \cdot X_j^{\min} \quad t \in \mathcal{T}, i \in \mathcal{I}_t \quad (3)$$

Given $\mathcal{J}_{t,i}$ the set of active tasks in (t, i) and $P^{\max}$ the power consumed by a single core at full utilization, Eq. (3)

TABLE I: Notation.

Name	Description
$\mathcal{T}$	Set of time slots. A slot $t \in \mathcal{T}$ is a period of unchanging renewable power levels.
$\mathcal{I}_t$	Set of intervals in time slot t . An interval $i \in \mathcal{I}_t$ is a period without change in the number of active tasks.
$\mathcal{J}$	Total tasks set.
$\mathcal{J}_i$	Set of active tasks at interval i .
$\mathcal{H}_i$	Subset of active tasks with utility larger than the minimal one at interval i .
$\mathcal{L}_i$	Subset of active tasks at minimal utility at interval i .
$l \in \mathcal{I}_k$	The most loaded interval during task k .
$N_l \in \mathbb{Z}_{\geq 0}$	The maximum load during task k .
$\mathcal{A}_{il}$	Artificial tasks set: tasks added to set $\mathcal{J}_i$ to get the maximal load N_l .
$x_{tj} \in \mathbb{R}_{\geq 0}$	Number of CPU cores (CPU usage) used by task j at slot t .
$U_j(x_j) \in \mathbb{R}_{\geq 0}$	Utility of task j at CPU usage x_j .
$U^{min} \in \mathbb{R}_{\geq 0}$	Minimal utility.
$X_j^{min} \in \mathbb{R}_{\geq 0}$	Value for x_j at minimal utility.
$F \in \mathbb{R}_{> 0}$	CPU core frequency (in GHz).
$f_{tj} \in \mathbb{R}_{\geq 0}$	Effective CPU frequency used to perform task j at slot t (in GHz) ($f_j = F \cdot x_j$).
$\theta_j \in \mathbb{R}_{\geq 0}$	Decay rate constant of $1 - U_j(x_j)$.
$\tau_j \in \mathbb{R}_{\geq 0}$	Value of $1 - U_j(x_j)$ at zero CPU usage ($x_j = 0$).
$R_t \in \mathbb{R}_{> 0}$	Renewable power (Watts) at slot t .
$S_{ti} \in \mathbb{R}_{\geq 0}$	Power (W) imported at slot-interval pair (t, i) .
$P \in \mathbb{R}_{\geq 0}$	Max core power consumption (W).
$P_{ti}^{min} \in \mathbb{R}_{\geq 0}$	Power (W) to assure U^{min} at slot-interval (t, i) .
$\lambda \in \mathbb{R}_{\geq 0}$	Lagrange multiplier for constraint (5b).
$\mu_j \in \mathbb{R}_{\geq 0}$	Lagrange multiplier for constraint (5c).

calculates the minimum power requirement at slot t and interval i . The resource allocation problem is therefore to assign $\{x_j\}_{j \in \mathcal{J}_{t,i}}$ such that $x_j \geq X_j^{min}$, $\forall j \in \mathcal{J}_{t,i}$, maximizing aggregated task utility. At each interval $i \in \mathcal{I}_t$, the active task set $\mathcal{J}_i$ is partitioned into $\mathcal{H}_i = \{j \in \mathcal{J}_i : U_j(x_{tj}) > U_j^{min}\}$ and $\mathcal{L}_i = \{j \in \mathcal{J}_i : U_j(x_{tj}) = U_j^{min}\}$. The minimum import required to satisfy all baseline allocations is:

$$S_{ti} = \max(0, P_{ti}^{min} - R_t), \quad t \in \mathcal{T}, i \in \mathcal{I}_t \quad (4)$$

Allocations $\{x_{tj}\}$ are updated only at the coarse scale $t \in \mathcal{T}$, while within each interval i , variations in due to task arrivals and departures are compensated by adjusting imports S_{ti} .

Oracle solution. The allocation problem, which we refer to as *Renewable Energy Aware Task Allocation (REATA)*, can be decomposed into $|\mathcal{T}|$ independent subproblems. Eq. (5) defines the *Oracle solution* for each slot $t \in \mathcal{T}$ as an offline convex optimization problem with full knowledge of the task set $\mathcal{J}$, including the arrivals and durations of all tasks.

$$\max_{\{\mathbf{x}\}} \sum_{j \in \mathcal{J}} (1 - \tau_j \cdot \exp(-\theta_j \cdot F \cdot x_{tj})) \quad (5a)$$

$$\text{s.t.} \quad \sum_{j \in \mathcal{J}_i} P \cdot x_{tj} \leq R_t + S_{ti}, \quad i \in \mathcal{I}_t, \quad (5b)$$

$$x_{tj} \geq X_j^{min}, \quad j \in \mathcal{J}, \quad (5c)$$

$$x_{tj} \in \mathbb{R}_{\geq 0}, \quad j \in \mathcal{J} \quad (5d)$$

$\mathbf{x} = x_{tj}$ are the decision variables representing the CPU allocation for task j at slot t , and S_{ti} denotes the imported power required to meet the minimum allocation demand. The objective function (5a) maximizes the aggregate utility of active tasks, subject to the minimum utility requirement (5c) and the power budget constraint (5b).

Online-REATA. To tackle the online version of the *REATA* problem, where tasks are assigned as they arrive, we devise a heuristic algorithm from applying the Karush-Kuhn-Tucker (KKT) conditions [14] to the formulation (5). Unlike the offline case, tasks' characteristics are not available in advance but revealed over time, and decisions are based only on current task states and predictions about future states. We decompose the oracle formulation into $|\mathcal{J}|$ subproblems, each solved upon the arrival of a task $k \in \mathcal{J}$. Here, the notation relating to slots $t \in \mathcal{T}$ will be omitted ($R_t \rightarrow R$, $S_{ti} \rightarrow S_i$).

Assuming that tasks $j \in \mathcal{J}_i \setminus k$ from other intervals are optimally allocated, and that constraints (5b) can be replaced with the tightest one, i.e., the one corresponding to the most loaded interval l during the task k , the problem reduces to Eq. (6):

$$\max_{\{x_k\}} U_k(x_k) \quad \text{s.t.} \quad \sum_{j \in \mathcal{J}_i \setminus k} x_j^* + x_k \leq \frac{R+S_l}{P}, \quad x_k \geq X_k^{min} \quad (6)$$

By convexity, KKT conditions yield the allocation rule:

$$x_k^* = \begin{cases} X_k^{min} & \text{if } \lambda > U'_k(X_k^{min}) \\ -\frac{1}{\theta_k F} \ln\left(\frac{\lambda}{\tau_k \theta_k F}\right) & \text{if } \lambda \leq U'_k(X_k^{min}) \end{cases} \quad (7)$$

where the dual variable λ is obtained from the constraint (5b):

$$\lambda = \exp\left(\frac{-F \left(\frac{R+S_l}{P} - \sum_{j \in \mathcal{L}_i} X_j^{min}\right) + \sum_{j \in \mathcal{H}_i} \frac{\ln(\tau_j \theta_j F)}{\theta_j}}{\sum_{j \in \mathcal{H}_i} \frac{1}{\theta_j}}\right) \quad (8)$$

The λ value computed as in Eq. (8) can be used in the allocation rule (7) to determine if the optimal allocation x_k^* of the task k is equal or larger than X_k^{min} , by replacing λ in (7) when $x_k^* > X_k^{min}$. The full mathematical derivation is described in Section I of the supplementary material.¹

The pseudocode of the heuristic is shown in Algorithm 1. At the arrival of a new task k , it is estimated whether the current interval i corresponds to the most loaded interval l for the time slot, based on the estimated maximal load $\hat{N}_l$. If not, artificial tasks $\mathcal{A}_{il}$ are generated to emulate the expected future load, ensuring that the computation of the dual variable λ in Eq. (8) reflects the maximum load scenario. The allocation x_k is then determined according to the rule in Eq. (7): if $\lambda > U'_k(X_k^{min})$, the task is assigned its minimum allocation X_k^{min} and added to $\mathcal{L}_i$; otherwise, it receives a higher allocation $x_k > X_k^{min}$ and is added to $\mathcal{H}_i$. The active set is updated accordingly as $\mathcal{J}_i = \mathcal{H}_i \cup \mathcal{L}_i$.

¹Available at: <https://hal.science/hal-05326830>

Algorithm 1: ONLINE-REATA: abstract task allocation

Data: $\mathcal{J}_{i-1}, \mathcal{H}_{i-1}, \mathcal{L}_{i-1}, \hat{N}_l, S_{i-1}, R$
 k : arriving task at interval i (slot t).
Result: Optimal allocation x_k for task k

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1 for each arriving task  $k$  do
2   if  $|\mathcal{J}_{i-1}| + 1 \geq \hat{N}_l$  then
3      $\hat{\mathcal{H}}_i \leftarrow \mathcal{H}_{i-1} \cup \{k\}$ ;
4   else
5     Generate artificial tasks  $\mathcal{A}_{il}$  based on  $\hat{N}_l$ ;
6      $\hat{\mathcal{H}}_i \leftarrow \mathcal{H}_{i-1} \cup \{k\} \cup \mathcal{A}_{il}$ ;
7   Compute  $\lambda$  using (8) with  $(\hat{\mathcal{H}}_i, \mathcal{L}_{i-1}, R, S_{i-1})$ ;
8   if  $\lambda > U'_k(X_k^{min})$  then
9      $x_k \leftarrow X_k^{min}$ ;
10     $\mathcal{L}_i \leftarrow \mathcal{L}_{i-1} \cup \{k\}$ ;
11  else
12     $x_k \leftarrow -\frac{1}{\theta_k F} \ln\left(\frac{\lambda}{\tau_k \theta_k F}\right)$ ;
13     $\mathcal{H}_i \leftarrow \mathcal{H}_{i-1} \cup \{k\}$ ;
14   $\mathcal{J}_i \leftarrow \mathcal{H}_i \cup \mathcal{L}_i$ ;
  
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IV. CASE STUDY

In this section we describe our case study, which corresponds to a grid-connected data center equipped with a hybrid energy system, including a microgrid with solar panels and wind turbines. The data center performs live video transcoding operations, an example of compressible tasks. A Lithium-Ion (LI) battery, well suited for cloud applications [3], is integrated into the system and charged exclusively when excess renewable energy is available, until its capacity is fully reached. Consequently, power imports can be supplied either by the battery or by the main grid, with the battery being prioritized for both import and export operations. A converter regulates transfers between AC and DC components [2] (see Fig. 1).

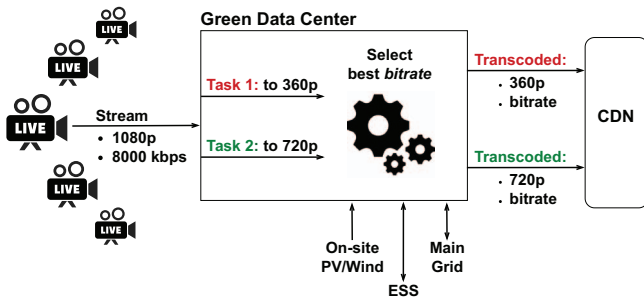


Fig. 1: Case study description. The 1080p, 8000 kbps video stream must be transcoded to 720p and 360p. The stream arrives at the data center, where the *Online-REATA* algorithm is applied to find the optimal bitrate based on the renewable energy available. The transcoded video streams are then delivered to end users via CDN.

The data center receives video streams from live broadcast channels and prepare them for the Content Distribution Network (CDN), which then delivers the transcoded streams to edge servers serving the end-users [15]. In accordance with guidelines [16], the broadcasting server originally encodes video streams at 1080p and 8000 kbps. We will refer to this resolution-bitrate (1080p, 8000 kbps) pair as the *original encoding profile*. This profile may not be suitable for users with low-resolution devices or slow connections; therefore, videos are transcoded into 360p and 720p to ensure playback on these devices. In this case study, we define a *task* as the *request to live-transcode* a channel's video stream at one of these two resolutions. Since different *transcoding profiles* (resolution-bitrate) require different CPU usage and energy resources (higher bitrates require more resources), renewable availability must be considered when choosing a profile.

CPU usage model. The computational effort for transcoding operations varies with the video content type (Table II) and the selected target profile. We obtained these values by transcoding, with *ffmpeg* [17], a video of fixed duration w for each content type, converting from the original profile to both target resolutions and sampled bitrates. For each operation, we measured the transcoding time c and, then, estimated the number of CPU cycles per second (i.e., frequency) required to transcode w seconds as $f = \frac{c \cdot F}{w}$, with F as the maximum CPU frequency. Then, we fitted a linear relation between the sampled bitrates r and its corresponding frequency f , obtaining coefficients m and n for each (content type, target resolution) pair, such that $r(f) = mf + n$.

Energy consumption model. For each transcoding task with a frequency f , the power consumption was estimated as $x \cdot P$, where P is the maximum power consumption of a core, and x is the task core usage calculated as $x = \frac{f}{F}$.

Utility model. The utility of a transcoding task is defined as the Quality of Experience (QoE), which depends on video content, target resolution, and bitrate. Following [15], QoE is modeled through the VQM metric as $U(r) = 1 - a \exp(-br)$, where r is the target bitrate. To relate this to the utility function in (1), we express r as the function of the frequency f derived above: considering $\tau = a \exp(-bn)$, $\theta = bm$, we obtain $U(r(f)) = 1 - \tau \exp(-\theta f)$. Table II reports the fitting parameters θ and τ , and Fig. 2 shows the resulting curves. Further details are provided in Section II of the supplementary material.¹

Real-time constraint. Upon the arrival of each task, Algorithm 1 simply applies Eq. (7), resulting in negligible latency. Frequency f ensures that transcoding is completed within the playback time of each fragment, avoiding additional delay.

V. EVALUATION

A. Experimental Settings

We evaluate *Online-REATA* over a **three-day scenario** (2021-06-15, 2021-06-16, and 2021-06-19), where both traffic intensity and renewable energy availability vary over time. All simulations were run on a 13th generation Intel Core i7-13850HX (2.2–5.3 GHz, 20 cores with 8 performance and 12

TABLE II: Test videos and type, τ and θ parameters by content type and device resolution.

Content Type	Resolution	τ	θ
Documentary Aspen, Snow Mountain	360p	4273873.86	1.162×10^{-8}
	720p	116.93	2.042×10^{-9}
Movie Old Town Cross	360p	4951.70	5.776×10^{-9}
	720p	4234.21	2.389×10^{-9}
Sport Touchdown Pass, Rush Field Cuts	360p	560.79	3.400×10^{-9}
	720p	73.18	1.046×10^{-9}

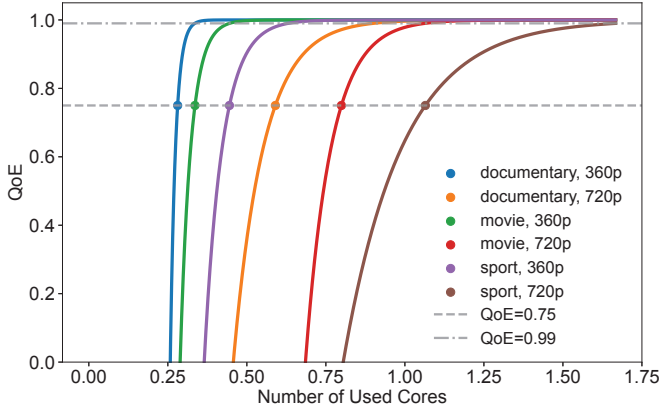


Fig. 2: Fitting curves of QoE vs. number of cores x ($x = \frac{f}{F}$). Assuming a maximum per-core frequency of 5.1 GHz, additional cores are allocated when a single core is insufficient.

efficiency cores), with 32 GB RAM, under Ubuntu 22.04 LTS.

Traffic Intensity. We rely on the Twitch traffic dataset published in [18], [19]. From this worldwide dataset, we extract *French-language live streams* to align user demand with renewable energy sources from the French grid in terms of timezone. Timestamps are adapted to UTC+2. To capture content diversity, we identify the most popular Twitch categories [20] (e.g., *Just Chatting*, *Minecraft*) and map them to three representative video classes: documentary (25.4%), movie (45.8%), and sports (28.8%), based on visual and dynamic similarity. Traffic intensity is defined as an estimate of maximum number of concurrent tasks per slot ($\tilde{N}_t$). In this work, we assume that this estimate is highly accurate (we use actual values), since many reliable techniques are available to forecast seasonal streaming patterns, such as SARIMA, Fourier analysis and recurrent neural networks (RNNs).

Renewable Availability. For the same three days, we obtained hourly wind and solar production data from the French grid [21]. These values were interpolated to capture finer fluctuations (15-minute time slots). To align renewable supply with the Twitch workload, we compute the daily renewable budget E_{24h} , i.e., the energy required to serve all tasks at a target QoE of 0.99. This budget is distributed across slots proportionally to the grid's renewable profile: even if the daily renewable budget

is theoretically sufficient, mismatches between demand and supply occur since the temporal distribution does not follow the workload's trend (see Fig. 3).

Benchmarks. We compare *Online-REATA* against three reference strategies: (1) *Benchmark-0.99*, which assigns each task the number of cores needed to reach a target QoE of 0.99 (maximum QoE), importing energy when renewables are insufficient. It provides the upper bound for QoE and import and a lower bound for export; (2) *Benchmark-0.75*, which assigns each task the allocation to achieve a target QoE of 0.75 (minimum QoE), providing a lower bound for export and an upper bound for import; and (3) *Benchmark-0.89*, which satisfies every task with a QoE of 0.89, the average QoE that *Online-REATA* achieves.

Data center and energy system. Battery charging and discharging efficiencies are set to $\eta_{ch} = \eta_{disch} = 0.9$, assuming a Depth of Discharge (DoD) = 0.8 [3]. DoD defines the usable portion of battery capacity [11] and it is set to avoid premature degradation. The battery prioritizes renewable energy exchange over any interaction with the grid. For the tests involving a battery, the system is initialized with the battery initially discharged, and charging and discharging processes of the battery are simulated. Converter efficiency is 1, battery degradation and idle power are neglected (the latter would add constant offset to dynamic consumption and it has been reduced by recent hardware efficiency improvements [1]). All cores are assumed homogeneous, with frequency F and power consumption P .

B. Results

We evaluate *Online-REATA* against different benchmarks by varying battery size and analyzing its impact on grid interactions, QoE, and the effective use of storage. First, we ran the tests without batteries. Fig. 4 reports, for each time slot of June 17, 2021, the reduction in total imports plus exports (i.e., interactions with the main grid) together with the average QoE achieved. Over the three-day simulation, *Online-REATA* reduces average daily imports by 52.76% relative to *Benchmark-0.99* and cuts exports by 21.62% compared to *Benchmark-0.75*, while ensuring an average QoE of 0.89.

Secondly, experiments are conducted by introducing batteries and varying their capacity in 500 Wh increments, ranging from 0 Wh to 5500 Wh. Figures 5 and 6 illustrate the results, where the performance of each algorithm is evaluated using the SSR and SCR metrics, as defined in [7]. SCR and SSR computation for the considered case study is discussed in Section III of the supplementary material.¹ We observed that increasing battery size past a certain point yields diminishing returns for SCR and SSR. This occurs because larger batteries take longer to reach the minimum DoD, delaying the discharge phase. This reduced utilization of stored renewable energy leads to lower effective SCR and SSR values, since all the discharged energy from the battery counts as renewable supply. Figures 5 and 6 confirm that, among all benchmarks, *Online-REATA* achieves the best trade-off between SCR and SSR, without requiring large battery storage, while maintaining an

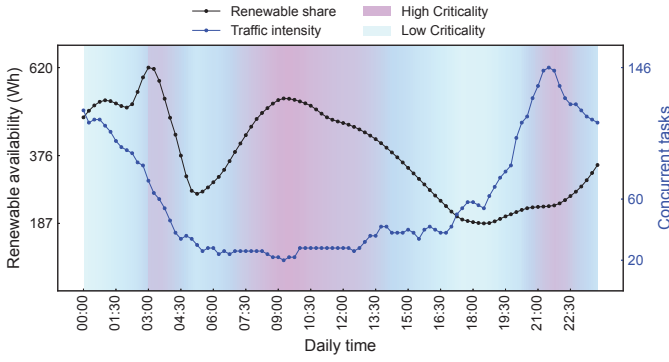


Fig. 3: Traffic and energy patterns at Day 3, 2021-06-17: Load trend (blue line) and energy profile (black line) during the simulated days. The heatmap indicates the criticality of each slot: the criticality is higher when the mismatch between resources and demand is accentuated.

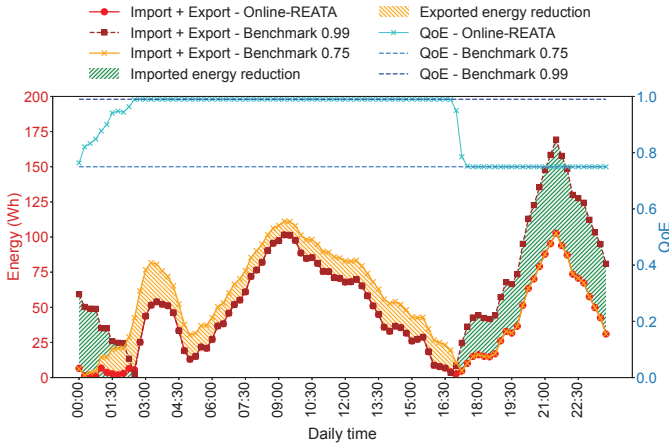


Fig. 4: The sum of the energy imported and exported at Day 3, 2021-06-17, by *Benchmark-0.99*, *Benchmark-0.75* and *Online-REATA* and the average QoE for each time slot are represented.

average QoE that is only about 10% below the maximum value. We also observe that *Online-REATA*, for an equivalent average QoE (0.89), is able to match the best SSR performance of *Benchmark-0.89* (≈ 0.97), while improving SCR by about 3.4% and doing so with a 35.3% smaller battery capacity (from 5035 Wh to 3257 Wh). Additionally, for an equal battery size (500 Wh), *Online-REATA* is able to deliver an SCR improvement of +11.24% while maintaining an SSR improvement of +7.57%. We underline that the high self-consumption observed in *Benchmark-0.99* is mainly due to the fact that the renewable availability was dimensioned according to the energy required to achieve the maximum QoE of 0.99. Despite this, in the absence of a battery, *Online-REATA* is able to almost match *Benchmark-0.99* in terms of SCR (only -0.62% difference), while significantly outperforming it in SSR, with an improvement of +23.18%. Moreover, for almost all battery sizes, *Online-REATA* maintains a higher SSR level compared to *Benchmark-0.75* (up to a battery size

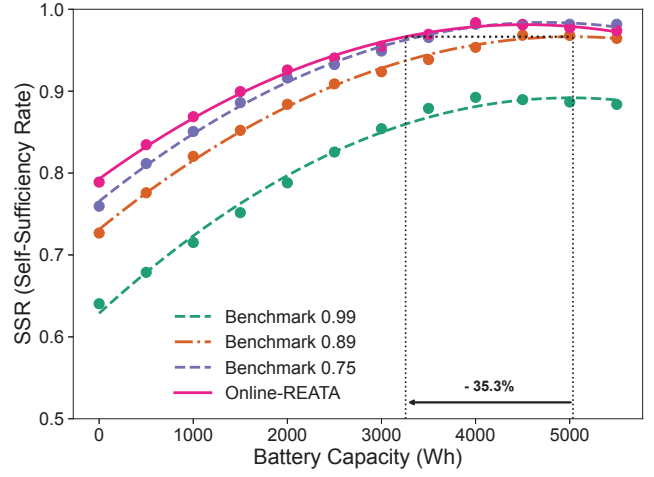


Fig. 5: SSR metric for each algorithm and battery capacity: it indicates the share of total load supplied by renewables.

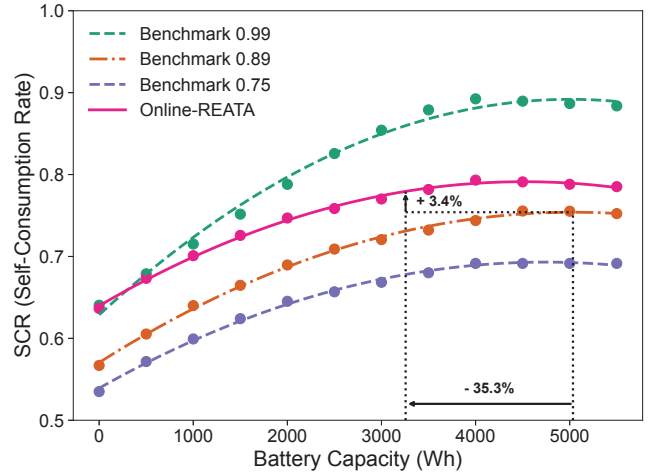


Fig. 6: SCR metric for each algorithm and battery capacity: it measures the share of renewable energy used over that available.

of ≈ 4000 Wh), and substantially increases SCR, achieving up to a +17.78% improvement with a 500 Wh battery while maintaining a higher average QoE.

VI. CONCLUSION

This work proposes an approach based on *compressible tasks* to reduce green data centers' reliance on the main grid while preserving QoE. The approach jointly addresses imports reduction, final QoE, grid stability, and suitability for delay-sensitive workloads. We also aimed to reduce the reliance on storage systems. These aspects, to the best of our knowledge, have not been addressed together in prior work. We developed the *Online-REATA* heuristic, and we evaluated it in the case study of live video transcoding: *Online-REATA* achieved up to $\approx 11\%$ higher SCR and $\approx 8\%$ higher SSR than constant allocation (*Benchmark-0.89*), while maintaining the same average QoE without additional dependence on

energy storage. The gains came from adjusting the bitrate, reducing the import from the main grid but also absorbing the energy surplus. Future work will extend the proposed approach to other compressible tasks (e.g., ML inference) and to distributed environments, including space-shifting of delay-tolerant compressible tasks across data centers.

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